CAMBRIDGE INTERNATIONAL EXAMINATIONS

Cambridge Ordinary Level

MARK SCHEME for the May/June 2015 series

5014 ENVIRONMENTAL MANGEMENT

5014/22 Paper 2, maximum raw mark 60

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1 (a) (i) peak harvest and demand coincide; so no drop in price / price may increase / more (money / profit); all crop sold / no crop wasted; have money for any household expense / food / hospital / school fees; have money for farm costs; standard of living; [2]

(ii) 58, 62, 80%;
All three for two marks. Two for one mark. [2]

(iii) as a control / idea explained / idea of comparison; [1]

(iv) cost (of acids); do not know how to use them / suspicious of using them; extra time / work and not worth extra seedlings; ref. to acid contamination of soil; ref. to contamination of water; ref. harm to humans; low / unavailability of acids; [2]

(b) (heavy) metal poisonous / toxic; ref. contamination of (soil) water; absorbed by plants; not broken down / excreted (by animals); so accumulates in (animal / plant / organism) body; builds up / passes) along food chain; kills top predators; bioaccumulation; biomagnification; [3]

(c) (this / resting) restores nutrients / minerals / named mineral / ref. clover etc.; crops grow better / less fertiliser needed; allows pests / diseases to die out / fewer pests; crops grow better / yield not decreased; [2]

(d) (i) to check for (error / reliability) / can find a mean / backup if one tray fails; [1]

(ii) comparing 3 to 1: (more / 2) trays so reliability (can be checked); can find a mean / back up if one tray fails; [3]

comparing 3 to 2: because result measured twice / at 20 and 25 days;

allows (greater chance) of full germination / more accurate estimation of germination rate;

(iii) size of tray; volume (amount) of soil; depth of soil; type soil; pH of soil; temperature; (volume / amount) of water; (amount) of light; type / variety of seed; seed age; [3]

(e) (i) orientation of line graph only (not if bar chart);

axes labelled;
plots; [4]

(ii) increase then plateaus / levels off / cuts off / evens out / eq.; [1]

(iii) do not plant more than 200 per tray / do plant (between 150 and 200); above this seeds wasted / no further yield of seedlings / no money wasted / eq.; [2]

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(f) (i) (25 °C–22 °C =) 3 °C; [1]

(ii) (weather / rain / temperature) conditions remain constant / eq.; temperature always warm enough / 15 °C–20 °C; sufficient water; [2]

(iii) spread of plant disease / fungal infection; surface run-off; soil erosion; loss of topsoil; nutrient leaching; spread of (water-related / eq.) disease; crop damage / loss; [3]

(g) (i) selective breeding; best plants have good alleles (accept genes) / eq.; best plants pass on good alleles (accept genes) to next generation / eq.; further detail; [2]

(ii) genetic engineering; transfer of genes (for high yield); further detail; cloning / tissue culture / micropropagation; [2]

2 (a) still rely on imports / depend on Mexico / money going to Mexico; greenhouse gases / named greenhouse gas; leading to global warming / greenhouse effect / eq.; (finite / unsustainable) supply; costs of supplying more electricity increase; acid rain; H.E.P. more land flooded / relocation; AVP; [4]

(b) very remote / far away; difficult terrain; too far for effective powerlines; few power plants / power stations far away; ref. to cost of powerlines; not many people live there / (so) low demand; [2]

(c) (i) systematic sampling;

at noted location, use of compass; layout transect line (with tape); from powerlines into forest; secure with pegs / eq.; placing quadrat at intervals; identify species with book; record number of different species (in quadrat); in notebook; repeat;

OR

random sampling;

detail of quadrat placing (throw quadrat / layout grid); under powerlines and in forest; identify species with book; record different species in quadrat; in table in notebook; repeat; AVP; [5]

(ii) repeat on other sites (when repeating keep variables constant) / compare with similar studies; [1]

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(d) (i) 0100–0400 (or within this range); [1]

(ii) getting dark; more home; cooking; AVP; [2]

(iii) cheaper to invest in reducing demand; sustainable reductions / better use of resources; new power plants not sustainable; less reliant on imports; cheaper electricity / spending less on electricity helps GDP / eq.; environmental reasons; human health reasons; AVP; [4]

(e) (i) by using diesel generators / solar panels / water wheel / wind turbine; [1]

(ii) each of three questions about small-scale generation; layout;

such as:
Do you know how to run a generator / do you have the money to buy a generator?
Do you know anything about small wind turbines / could you pay for one?
Have you seen small solar panels working? [4]

AVP = Alternative Valid Point.

[Total: 60]

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